

PROJECT REVIEW

Banking Transaction Management System

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Project Type	Java / SQL
Technology Stack	Java, Spring Boot, Spring Data JPA, MySQL, REST API, Maven, Postman

Abstract

A banking backend application that manages users, accounts and financial transactions such as deposits, withdrawals and transfers. The system focuses on safe transaction processing and database consistency.

Introduction

Banking systems must process financial operations accurately even when multiple requests happen at the same time. A backend application with transactional database operations helps maintain correct account balances.

Problem Statement

Concurrent transactions can cause race conditions, incorrect balances or partial updates if operations are not handled atomically.

Solution for the Problem Statement

The system uses Java and Spring Boot REST APIs with Spring Data JPA and MySQL. Transactional processing ensures related database operations succeed or fail together. Concurrency control and account-level locking help protect balances during simultaneous operations.

Conclusion

The project demonstrates important backend concepts such as REST APIs, SQL database design, JPA relationships, transactions, concurrency control, validation and exception handling.

